

Institute of Actuaries of India

**Subject ST6 – Finance and
Investment B**

September 2016 Examination

INDICATIVE SOLUTION

Solution 1:

i. Type of longevity hedges are

- a. Longevity swap
- b. Longevity bonds
- c. Principal at risk
- d. Survivor cap

Further description of each of the hedge would be required.

[5]

ii. Value of initial swap is shown as difference of B-A wh

			A	B
			PV	Regulatory
			Discount	basis
			factor	cashflow
Interest rate			Best estimate	
1	5%	0.95238095	100000	110000
2	5%	0.90702948	100000	110000
3	5%	0.8638376	100000	110000
4	5%	0.82270247	100000	110000
5	5%	0.78352617	100000	110000

Which is 43,295.00

[5]

iii. The value of fall in the capital requirement is difference between C-B

			B	C	
			PV	Regulatory	
			Discount	basis	
			factor	cashflow	1/200
Interest rate					payouts
					Loss given default
1	5%	0.95238095	110000	165000	65000
2	5%	0.90702948	110000	165000	65000
3	5%	0.8638376	110000	165000	65000
4	5%	0.82270247	110000	165000	65000
5	5%	0.78352617	110000	165000	65000

value of fall in
in capital
requirement 238,121

[4]

- iv. The counterparty providing swap might default leading to insurance company exposed the risk of shortfall. There is concern that if the counterparty has other clients in the market hence sudden increase in the longevity might lead to failure of the reinsurer. The concern can be reduced by the way of collateral or retaining sufficient capital to replace the longevity swap as risk capital.

[4]

- v. Collateral value is calculated in the table below

			B	C			
PV			476,242	714,364	16,336		
Discount			Regulatory	1/200	Loss given	PD	=PD*LGF
Interest rate	factor		basis	payouts	default		
1	5%	0.95238095	cashflow	110000	65000	2%	1300
2	5%	0.90702948		110000	65000	4%	2600
3	5%	0.8638376		110000	65000	6%	3900
4	5%	0.82270247		110000	65000	8%	5200
5	5%	0.78352617		110000	65000	10%	6500
							19500

The collateral = 16,336

[6]

[24 Marks]

Solution 2:

i) Warrants

Warrants grant the holder the right - but not the obligation - to buy the underlying asset (stock) at a specific future date. Unlike options, warrants are issued by a company rather than an investor, and are offered to holders of company bonds and preferred stock. A warrant allows an investor to purchase (call warrant) or sell shares (put warrant) of a company's stock at a certain price, often much higher than the price at the time the warrant was issued. A company issuing a warrant can improve the yield on its bonds, and providing an investor the option of purchasing or selling stock later on often reduces the interest rate or dividend that the investor will receive.

[2]

ii) Convertible bonds

This is a type of bond that allows its holder to convert the bond into equity (stock) in the underlying company. The derivative still has the features of a bond, such as a coupon and a maturity date, but also contains the conversion rate and price in which the bond can be exchanged. These bonds are generally considered safer than common stock because of its bond provisions, and can be used in a convertible arbitrage strategy (simultaneous purchase of a company's convertible bond and shorting of the company's stock). Convertible bonds come in many variations, including vanillas, exchangeables, mandatory exchangeables, contingents, hybrids and reverse convertibles.

[2]

iii) Property derivative

A property derivative is a financial derivative whose value is derived from the value of an underlying real estate asset. In practice, because real estate assets fall victim to market inefficiencies and are hard to accurately price, property derivative contracts are typically written based on a real estate property index. In turn, the real estate property index attempts to aggregate real estate market information to provide a more accurate representation of underlying real estate asset performance. Trading or taking positions in property derivatives is also known as synthetic real estate.

Property derivatives usually take the form of a total return swap, forward contract, futures, or can adopt a funded format where the property derivative is embedded into a bond or note structure. Under the total return swap or forward contract the parties will usually take contrary positions on the price movements of a property index.

[2]

[6 Marks]

Solution 3:

i)

- Reputation of the builder including past completed flats.
- The property market current situation
- Price volatility of construction material
- Value of the property on which construction has been done
- Level of presale

Any other relevant factor is acceptable as the relevant solution.

[4]

ii) Result of the calculation

Percent of Principle	10%		Present value	1,085,473
amount	1,000,000		Allowance for default	85,473
Rate of interest	6%			
spread	2.50%			
Payout	Start of year	Principal	interest	total
1	1,000,000	100,000	85,000	185,000
2	900,000	100,000	76,500	176,500
3	800,000	100,000	68,000	168,000
4	700,000	100,000	59,500	159,500
5	600,000	600,000	51,000	651,000

[6]

iii) MV at end of 2 years

Current interest	2%
start of year 3 want to surrender	MV at start of year 3
MV	931,464
1	168,000
2	159,500
3	651,000

MV: 931,464

Payment is 800,000 hence loss is 131,464.

[4]

iv) Value of the cashflow discounted at inception yield

Current interest	6%	
start of year 3 want to surrender	MV at start of year 3	
MV		847,037
1	168,000	
2	159,500	
3	651,000	

Value of new mortgage

Interest rate charged 2% + 2.5%

Principle	800,000		<u>MV@6%</u>	767,923.86
	Principal	interest	total	
1	-	36,000	36,000	
2	-	36,000	36,000	
3	800,000	36,000	836,000	

Fine: $847,037 - 767,923 = 79,113$

Total payment = $800,000 + 79,113 = 879,113$

Shortfall = $931,464 - 879,113 = 52,350$

[6]

[20 Marks]

Solution 4:

- i. Ito's formula (or lemma) forms the basic extension of differential calculus to variables which are stochastic in nature.

It is used to derive stochastic differential equations (SDEs) for valuing derivatives whose payoffs depend on the evolution of a stochastic process.

Using suitable boundary conditions, these SDEs can then be solved analytically or numerically to give derivative prices and sensitivities.

[1]

- ii. A martingale X_t is a stochastic process under a probability measure $\mathbf{P}$, for which all expected future values of X , conditional on a known history up to time t , are equal to its current value at time t , i.e. $E_P[X_u | \mathcal{F}_t] = X_t$ for all $u > t$.

The martingale must also satisfy the boundedness condition $E_P[|X_t|] < \infty$ for all t .

[1]

iii.

$$\frac{\partial Y_t}{\partial t} = -1, \frac{\partial Y_t}{\partial W_t} = 2W_t \text{ and } \frac{\partial^2 Y_t}{\partial W_t^2} = 2.$$

Using Ito's Lemma, we get

$$dY_t = \left[-1 + \frac{1}{2} \cdot 2 \right] dt + 2W_t dW_t = 2W_t dW_t.$$

This is driftless and hence a martingale.

[2]

iv.

$$\frac{\partial Z_t}{\partial t} = -6W_t^2 + 2at, \frac{\partial Z_t}{\partial W_t} = 4W_t^3 - 12W_t t \text{ and } \frac{\partial^2 Z_t}{\partial W_t^2} = 12W_t^2 - 12t.$$

Using Ito's Lemma, we get

$$\begin{aligned} dZ_t &= \left[-6W_t^2 + 2at + \frac{1}{2}(12W_t^2 - 12t) \right] dt + [4W_t^3 - 12W_t t] dW_t \\ &= (2at - 6t)dt + (4W_t^3 - 12W_t t) dW_t. \end{aligned}$$

This is driftless if and only if $a = 3$.

[3]**[7 Marks]**

Solution 5:

- i. Differentiate the Put Call Parity twice. [2]
- ii. Vanilla is greater than discrete greater than continuous [3]
- iii. In general, it will be lower since the averaging process reduces overall variance in the risk neutral measure. [2]
- iv. The asset is drift less so when the American pays off there is precisely a 50% chance (in the risk neutral measure) that the European will pay off too. So the European is worth half as much as the American. [2]
- v. Price goes down [3]
- vi. This follows from Put Call parity and the fact the forward with this strike has zero value [3]
- vii. We know that $C = P + S - D - PV(X)$. Now, $S - D = S e^{-qT}$ because the stock price would be $S e^{-qT}$ today to reach the same terminal price if there were no dividends. [3]
- viii. The probability for $S_T \geq X$ equals that for $\ln(S_T/S) \geq \ln(X/S)$. Now, this event occurs with probability

$$1 - N\left(\frac{\ln\left(\frac{X}{S}\right) - \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}\right) = N\left(-\frac{\ln\left(\frac{X}{S}\right) - \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}\right) = N\left(-\frac{\ln\left(\frac{S}{X}\right) + \left(r - \frac{\sigma^2}{2}\right)T}{\sigma\sqrt{T}}\right)$$

Multiply this by e^{-rT} to get the desired result.

[5]

[23 Marks]

Solution 6:

- i. The price tree takes the values 160 at time 0, 240 or 80 at time 1, 360 or 120 or 40 at time 2 and 540 or 180 or 60 or 20 at time 3. The value of p is 0.7 and that of $(1 - p) = 0.3$. [4]
- ii. \$85 [4]
- iii. \$85.07 [4]
- iv. \$21.87 [3]
- [15 Marks]

Solution 7:

Let S be the stock price immediately before an ex-dividend date, let S' be the price immediately after an ex-dividend date and let D be the amount of dividend.

S' should be $S - D$ within so short a time interval.

So exercise immediately after an ex-dividend date fetches $X - S' = X - S + D$ and exercise immediately before an ex-dividend date fetches $X - S$.

The interest gained from $X - S$ in such a short period of time can be ignoring b .

The late exercise strategy clearly dominates.

[5 Marks]
